Lesson 6 Teacher's Key

TIC-80

INSTRUCTOR ONLY - do not distribute to students

Level 1 - Complete Worked Solution

One new counter variable, decremented inside the existing collision check, and a print() call gated on it reaching zero:

```
enemies={}
for enemy_count=0,4 do
 table.insert(enemies,{ex=20+enemy_count*20,ey=10,alive=true})
end
enemies_remaining=5
bullets={}

function TIC()
 cls(0)
 if btn(3) then x=x+1 end
 if btnp(4) then
  table.insert(bullets,{bx=x,by=y})
 end
 for i=#bullets,1,-1 do
  local b=bullets[i]
  b.by=b.by-2
  local hit=false
  for j=1,#enemies do
   local e=enemies[j]
   if e.alive and b.bx<e.ex+8 and b.bx+8>e.ex and b.by<e.ey+8 and
b.by+8>e.ey then
    e.alive=false
    enemies_remaining=enemies_remaining-1
    hit=true
   end
  end
  if hit then
   table.remove(bullets,i)
  elseif b.by<0 then
   table.remove(bullets,i)
  else
   spr(2,b.bx,b.by,0)
  end
 end
 for j=1,#enemies do
  local e=enemies[j]
  if e.alive then spr(3,e.ex,e.ey,0) end
 end
```

```
 spr(1,x,y,0)
 if enemies_remaining<=0 then
  print("You Win!",84,60,12)
 end
end
```

Level 2 - Complete Worked Solution

enemies_remaining is only ever decremented inside the same e.alive and ... guarded branch that already prevents a single enemy from being counted twice - so the win text appears exactly once it's actually earned, and stays showing from then on since the check is <= 0, not == 0.

Level 3 - All Worked Solutions

Every stretch-goal idea from the handout, fully worked - not just one. Useful for validating each idea actually works before class, and as a ready reference no matter which one a student ends up picking.

Sound: victory sound, once

Import the provided sfx_win.wav into a new SFX slot rather than designing a new effect, and add a has_won flag so it plays exactly once, not every frame the text is on screen:

```
has_won=false
...
 if enemies_remaining<=0 then
  if not has_won then
   has_won=true
   sfx(3)
  end
  print("You Win!",84,60,12)
 end
```

Visual: make it stand out

Draw a filled rectangle behind the text first so it's readable against any background:

```
 if enemies_remaining<=0 then
  rect(74,52,80,16,12)
  print("You Win!",84,60,15)
 end
```

Structure: one counter, one place to change it

enemies_remaining is declared once at the top with the rest of the game state, and only ever changed from inside the collision check - the same principle already used for enemies_alive.

Complete Level 3 solution (all 3 goals combined)

Everything above combined into the one TIC() function:

```
enemies={}
for enemy_count=0,4 do
 table.insert(enemies,{ex=20+enemy_count*20,ey=10,alive=true})
end
enemies_remaining=5
has_won=false
bullets={}

function TIC()
```

```
 cls(0)
 if btn(3) then x=x+1 end
 if btnp(4) then
  table.insert(bullets,{bx=x,by=y})
 end
 for i=#bullets,1,-1 do
  local b=bullets[i]
  b.by=b.by-2
  local hit=false
  for j=1,#enemies do
   local e=enemies[j]
   if e.alive and b.bx<e.ex+8 and b.bx+8>e.ex and b.by<e.ey+8 and
b.by+8>e.ey then
    e.alive=false
    enemies_remaining=enemies_remaining-1
    hit=true
   end
  end
  if hit then
   table.remove(bullets,i)
  elseif b.by<0 then
   table.remove(bullets,i)
  else
   spr(2,b.bx,b.by,0)
  end
 end
 for j=1,#enemies do
  local e=enemies[j]
  if e.alive then spr(3,e.ex,e.ey,0) end
 end
 spr(1,x,y,0)
 if enemies_remaining<=0 then
  if not has_won then
   has_won=true
   sfx(3)
  end
  rect(74,52,80,16,12)
  print("You Win!",84,60,15)
 end
end
```